MTBBench: A Multimodal Sequential Clinical Decision-Making Benchmark in Oncology (NeurIPS 2025)

github.com/bunnelab/MTBBench huggingface.co/datasets/EeshaanJain/MTBBench

MTBBench is a **benchmark for evaluating AI agents in oncology** by testing models in a **multimodal, longitudinal, and agentic setting**.

The benchmark has **two main tracks:** 1) the **multimodal track** based on 26 HANCOCK cases and includes 390 question-answer pairs with H&E slides, IHC images, hematology reports and clinical outcomes and 2) **longitudinal track** based on 40 MSK-CHORD cases and includes 183 questions about outcome, recurrence, and treatment progression over time.

Key tasks:
1. **Multimodal reasoning**: Can the model combine pathology images, IHC, hematology, genomics, reports
2. **Longitudinal reasoning**: Can the model reason over patient history and changing clinical events?
3. **Tool use:** Can the model decide when to retrieve external information or call specialized tools?

Strengths:
MTBBench moves **beyond static medical QA** and evaluates models in a setting closer to actual oncology workflows. It is **clinician-validated** and includes both **multimodal and longitudinal patient data**. Another strength is the **agentic setup** - models are not given all information at once but must decide what and when to inspect.

Limitations:
MTBBench reflects clinical decision-making in oncology, but the **number of validated cases is limited (26 and 40)**.

Research Proposal

- expand **from general to medical LLMs** (experiment 1);
- shift **from plain accuracy estimation towards evaluation of clinically important behavior** like **uncertainty awareness** (experiment 2a, 2b) and **ability to selectively use external evidence** (experiment 3).

Experiment 1: General models vs medical LLMs

I added **Llama 3.1 and its medically tuned counterpart, Meditron**, generously shared by [Annie Hartley's group](), to the benchmark and compared their accuracy. For this minimal experiment, I used small 8B models. The experiment was designed as a text-only pilot rather than a full multimodal evaluation - 156 questions remained after excluding 234 image-based questions from the full 390-question benchmark. The retained categories were coagulation (37), renal (27), hematology (26), survival (26), recurrence (26), electrolytes (13), and glucose (1). Preliminary results suggest that **Meditron outperformed Llama 3.1** (Figure 1), achieving 64% accuracy versus 58% with large gains in prognosis (+8 pp) and hematology (+5 pp). Medical fine-tuning may be beneficial, especially for clinically specific reasoning.

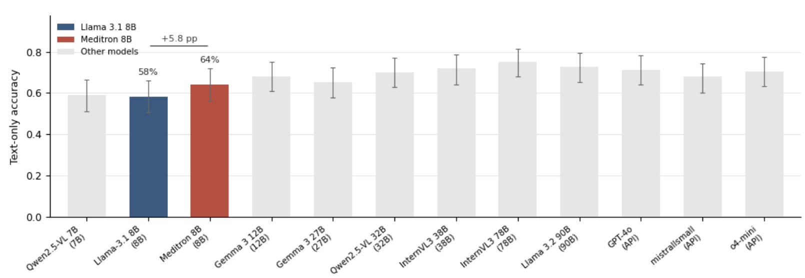

Figure 1. Model accuracy.
Models are ordered by increasing size. Previously benchmarked models are shown in gray; Llama 3.1 8B and Meditron 8B are highlighted in color.

Experiment 2a: Do models recognise when they are wrong?

I added confidence logging in two forms: log-probability of the selected answer and self-reported confidence from the model response. Then I compared confidence with actual accuracy. **Both models were overconfident, especially self-reported** – 0.88-0.94 with actual accuracies of 0.66-0.68; log-probability confidence was more conservative (Figure 2), but calibration error remained high (ECE 0.2-0.23).

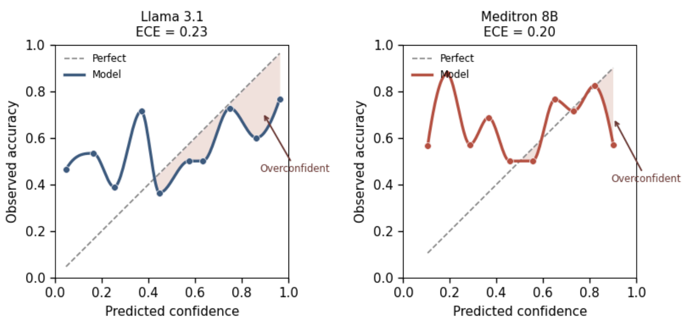

Figure 2. Calibration plot.
Reliability for Llama and Meditron using predicted confidence (logprobs) versus observed accuracy. Meditron 8B slightly better with lower ECE.

Experiment 2b: Could confidence predict errors?

Confidence was **not a universal error detector** (Figure 3), but for Llama it showed a useful signal in hematology and electrolytes (AUC > 0.7). This suggests that, **in some categories, low confidence could be used as an auto-trigger for additional retrieval or search**.

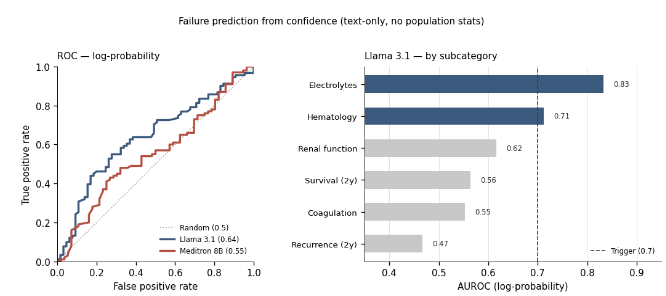

Figure 3. Confidence for failure prediction.
Logprobs confidence weakly predicted errors overall, with better performance for Llama 3.1 than Meditron 8B. For Llama, the signal was strongest in electrolytes and hematology, exceeding the AUROC >0.7 threshold.

Experiment 3: Selective usage of external evidence

For prognosis questions I asked models to i) stage the patient, ii) match the case to the TCGA/SEER cohort statistics, including survival and recurrence rates, and iii) inject this into the prompt together with a naive rule: "outcome is favorable if the cohort rate is clearly above chance (pre-specified heuristic >= 50/50 + 10 pp)" (Figure 4).

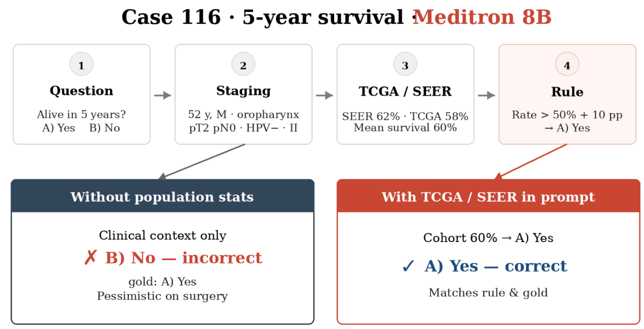

Figure 4. Prompting the model with population data.
Survival statistics shifts the prediction from an incorrect pessimistic answer to the correct 5-year survival label.

In prognosis questions **a naive cohort-rule baseline achieved 67.3% accuracy**, **Llama remained below (63.5%), while Meditron succeeded this threshold (69.2%).**
Importantly, Llama seemed to follow population recommendations almost blindly, matching it on 92% vs 75% for Meditron. Difference became clearest when the models rejected naive rule - Llama was correct only in 25% vs 54% (Figure 5).

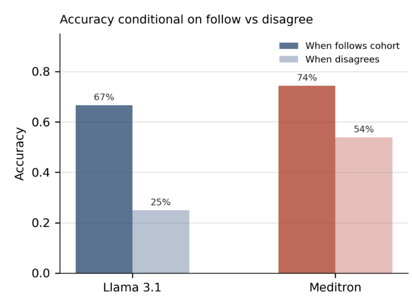

Figure 5. Accuracy conditional on cohort agreement.
Both Llama 3.1 and Meditron are more accurate when their predictions follow cohort-level survival statistics; the drop under disagreement is larger for Llama 3.1.

Conclusion. MTBBench is a strong benchmark that evaluates medical AI in a realistic setting. My extensions shift it from measuring accuracy toward domain adaptation and evaluation of clinically important behavior: uncertainty awareness and ability to selectively use external evidence.